Rajagopal Somaskandan

+91 7200288242 | somaskandan931@gmail.com
linkedin.com/in/somaskandan-rajagopal | github.com/Somaskandan931 | rajagopal-somaskandan.netlify.app

Summary

IEEE-published Machine Learning Engineer and national hackathon winner (Industry Innovation '26) specializing in end-to-end AI systems, from fraud detection pipelines to RAG-based career platforms. Proficient in PyTorch, FastAPI, and LLM fine-tuning with production deployment experience via Docker and cloud-ready microservices. Seeking an entry-level ML/AI engineering role to build scalable, explainable AI systems.

Education

Sathyabama Institute of Science and Technology 2025 – 2027 (In Progress)
M.Sc. Computer Science (AI Specialization)

Sathyabama Institute of Science and Technology 2025
B.Sc. Computer Science (AI Specialization) – ***First Class with Distinction***

Technical Skills

Languages: Python, Java, SQL
ML & AI: PyTorch, TensorFlow, Scikit-learn, LightGBM, XGBoost, SHAP, MLflow
NLP & Retrieval: Sentence-Transformers, FAISS, ChromaDB, LangChain, RAG, LLM Fine-Tuning (LLaMA)
Computer Vision & 3D: OpenCV, COLMAP, Gaussian Splatting, FFmpeg
Backend & Data: FastAPI, Pandas, NumPy, Selenium, PostgreSQL, MongoDB
DevOps & Cloud: Docker, CI/CD, AWS Fundamentals, Git, Github
Frontend: React, Tailwind CSS, Streamlit

Work Experience

Production AI Developer (Intern) Mar 2026 – Jun 2026
InnoXR Labs Pvt. Ltd. *Chennai, India*

- Developed an applied **machine learning pipeline** for single-video **3D reconstruction** (MonoSplat) using **COLMAP** for camera pose estimation and **Gaussian Splatting** in **PyTorch**.
- Worked on **AI** system design, building a hybrid **CPU-GPU data preprocessing pipeline** with **FFmpeg** for cloud-based execution.
- Deployed a scalable **FastAPI** backend with async job handling and a **Three.js** viewer for end-to-end 3D visualization, within an immersive technology startup environment.

Data Analyst Intern Jul 2024 – Oct 2024
India Advocacy *Chennai, India*

- Built Java and Selenium **ETL pipelines** to scrape and process **1M+ records**; performed **data preprocessing and cleaning** to transform raw data into **15–20k structured datasets** through multi-stage validation.

Additional Experience

Security Trooper – National Service Jan 2020 – Jan 2022
Singapore Armed Forces *Singapore*

- Completed mandatory national service; developed operational discipline, protocol compliance, and structured data reporting under time-critical conditions.
- Supported COVID-19 response initiatives through structured data collection and reporting under time-sensitive conditions.

Sales Associate (Part-Time) Feb 2022 – Aug 2022
Bata *Singapore*

- Managed customer interactions, billing, and inventory in a high-volume retail environment.

Projects

FinGuard Pro: Real-Time Fraud Detection & Compliance System IEEE ICAAIC 2025

- Developed a real-time UPI fraud detection pipeline trained on **100,000 synthetic transactions** using **25+ engineered features**, including geo-deviation scores and amount-to-average behavioral ratios.
- Evaluated **XGBoost**, **LightGBM**, and **CatBoost** using **Optuna TPE hyperparameter tuning**; selected **XGBoost** achieving **94% recall**, **ROC-AUC 0.9951**, and **F1-score 0.57**.
- Integrated **SHAP TreeExplainer** for local/global explainability and built a **fuzzy AML screening engine** using **RapidFuzz** over a **5,000-entity watchlist** with rule-based detection for structuring, round-tripping, and velocity anomalies.
- Built a dual-dashboard architecture using **FastAPI** and **Streamlit** with real-time alerts, SHAP visualizations, and automated PDF/ZIP compliance report generation, achieving sub-**3s inference latency**.

CareerGenie: AI Career Platform Hackathon Winner, Industry Innovation '26

- Built an end-to-end **RAG-based career platform** using **MiniLM embeddings** and **ChromaDB** for semantic job matching beyond keyword-based retrieval.
- Designed a **multi-agent orchestration system** with specialized Resume, Job, Interview, Roadmap, and Advisor agents, including an **Agent Debate pipeline** (proposer–critic–synthesizer) to improve response reliability and reduce hallucinations.
- Implemented a **Learning-to-Rank recommendation engine** using **Bayesian Personalized Ranking (BPR)** trained on click, application, and offer signals with weighted semantic and skill-based ranking.
- Integrated a multi-provider LLM fallback pipeline (**Groq** → **Anthropic**) to improve response stability and reduce degraded-provider latency during inference.
- Delivered a full-stack deployment using **React + Tailwind**, **FastAPI**, **Netlify**, and **ngrok**, supporting ATS scoring, resume rewriting, mock interviews, and personalized learning roadmaps.

SourceUp: Constraint-Aware Supplier Ranking System Final Year Research Project, 2026

- Built a **learning-to-rank** supplier discovery pipeline over a weak-supervision benchmark of **˜23,000 supplier–query pairs** (309 queries, **1.27M supplier records**) scraped from GlobalSources, IndiaMart, and TradeIndia using a Java/Selenium pipeline with adaptive fallback selectors.
- Trained a **LightGBM LambdaRank** model incorporating cost, delivery, reliability, and fairness constraints; evaluated against BM25, rule-based, and random baselines across **NDCG@5/10**, **MAP**, and **MRR** metrics with ablation and label-noise sensitivity analysis.
- Integrated **SHAP** feature-importance explainability and a fairness audit layer to quantify supplier-type representation disparities across ranked outputs.
- Implemented **SBERT + FAISS** semantic retrieval with hard-constraint filtering (budget, lead time, MOQ) to ensure business-rule compliance prior to ranking; pipeline paper targeting **IEEE conference submission**.

S.A.F.E: Privacy-Preserving Crowd Anomaly Detection Design Thinking Project, 2026

- Developed a privacy-preserving crowd anomaly detection system using **Farneback optical flow** and a **2×3 zone-grid architecture** to extract density, velocity, and directional variance metrics without tracking individuals.
- Conducted cross-model evaluation across **5 unsupervised anomaly detection models** (MAD, Z-Score, One-Class SVM, Isolation Forest, and LSTM Autoencoder) trained on **15,928 ETH/UCY records** and validated on **13,805 Zara1/Zara2 records**.
- Selected **Median Absolute Deviation (MAD)** as the primary detector after cross-domain testing on the Mall dataset, achieving **ROC-AUC 0.8391** and reducing false positives by **40–60%** through adaptive camera-shake filtering.
- Built a **FastAPI** backend and **Streamlit** dashboard with explainable real-time alerts and zone-level reasoning for crowd safety monitoring.

Publications & Conferences

FinGuard Pro: Explainable AI for Financial Fraud Detection | *ICAAIC 2025* ISBN: 979-8-3315-6587-9

Achievements & Hackathons

Industry Innovation Hackathon | *Sathyabama Institute of Science and Technology – National Level* Mar 2026

- Won **Best Industry Innovation Award** at Industry Innovation '26 competing against **63+ teams**.

Origin 24-Hour Hackathon | *SIMATS Engineering, Chennai* Apr 2026

- Placed **4th** at Origin 24-Hour Hackathon.

Certifications

AI & ML | *Space Zee Technologies* Jul 2024
Python Core | *Space Zee Technologies* Feb 2024